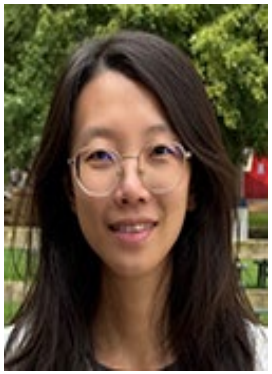
Measurement Error Models

- Donna Spiegelman
- Molin Wang
- Jingyu Cui
- Raymond Carroll

Who We Are

Donna

Molin



Jingyu

Raymond

Speakers

- Donna Spiegelman, Yale SPH, formerly Harvard SPH, over 800 papers, over 40 years in the field
- Molin Wang, Harvard SPH, Epi, Biostat, many papers, h-index 71, i10-index 220
- Jingyu Cui, Yale SPH, Postdoc with Donna
- Raymond J. Carroll, TAMU, Stat
- Good, brief biosketches via AI, e.g., coPilot

Outline

- Introduction #1 (Carroll)
- Introduction #2 (Spiegelman)
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Introduction

Outline

- Measurement Error Examples
- Some general results

The Triple Whammy of Measurement Error

- **Bias** in parameter estimation
- **Loss of Power** for detecting signals
- **Masking** the features of the data: measurement error causes the model of the observed data to change

Why Special Methods Are Needed — Cont.

- Better to replace W by a function of W, namely $E(X|W)$, since
$$X = E(X|W) + V,$$
where $E(X|W)$ and V are uncorrelated (**This is the regression property for residuals**)

Why Special Methods Are Needed — Cont.

- **New Model**: $E(Y|W) = \beta_0 + \beta_1 E(X|W)$ so

$$Y = \beta_0 + \beta_1 E(X|W) + (\epsilon + \beta_1 V)$$

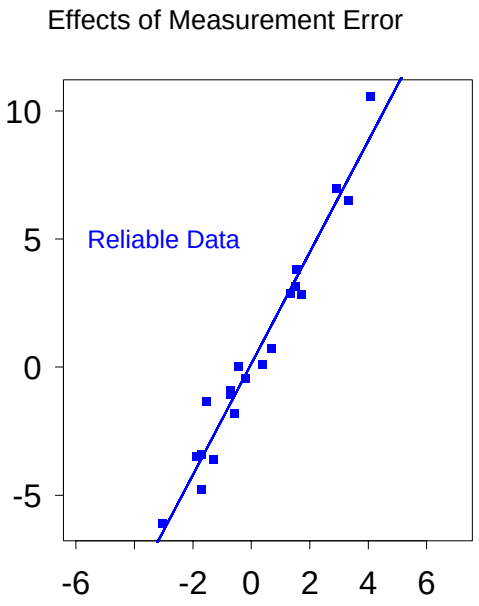
  - special case of "**regression calibration**"
  - leads to unbiased estimates
  - Notice the inflation of variance: $\sigma_\epsilon^2 + \beta_1^2 \sigma_V^2$.

Examples Of Measurement Error Problems

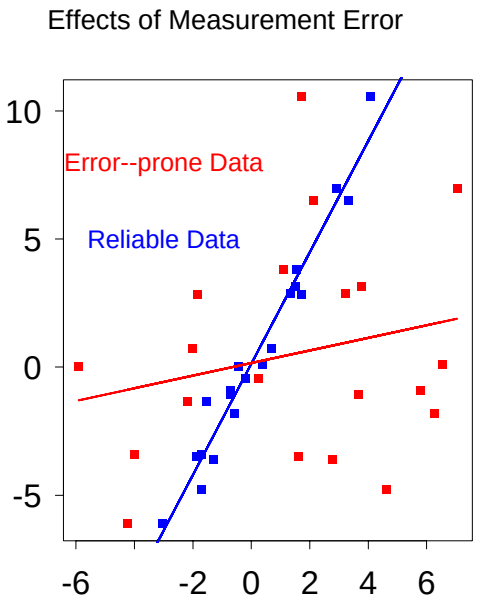
- Measuring usual nutrient intake
- Measuring Systolic Blood Pressure
- Measuring Radiation Dose from the Nevada Test Site
- Measuring Exposure to arsenic in drinking water, dust in the workplace, radon gas in the home, and other environmental hazards

Simulation Study

- Generate $X_1, \ldots, X_{50}$, IID $\mathsf{N}(0, 1)$
- Generate $Y_i = \beta_0 + \beta_x X_i + \epsilon_i$
  - ϵ_i IID $\mathsf{N}(0, 1/9)$
  - $\beta_0 = 0$
  - $\beta_x = 1$
- Generate $U_1, \ldots, U_{50}$, IID $\mathsf{N}(0, 1)$
- Set $W_i = X_i + U_i$
- Regress Y on X and Y on W and compare



True Data Without Measurement Error



Observed Data With Measurement Error

Multiple Linear Regression

- **Triple Whammy**
  - **Bias**
  - **Increased variance** and loss of power
  - **Loss of features**

- In multiple linear regression, the bias can take unusual forms

- This can lead to **invalidity of hypothesis testing**

Multiple Linear Regression With Error

- **Model**

$$
\begin{aligned}
Y &= \beta_0 + \beta_z^{\mathsf{T}} Z + \beta_x^{\mathsf{T}} X + \epsilon \\
W &= X + U
\end{aligned}
$$

- **Bias** Regressing Y on Z and W estimates

$$
\begin{pmatrix} \beta_{z*} \\ \beta_{x*} \end{pmatrix} = \Lambda \begin{pmatrix} \beta_z \\ \beta_x \end{pmatrix} \qquad \left[\neq \begin{pmatrix} \beta_z \\ \beta_x \end{pmatrix} \right]
$$

Multiple Linear Regression With Error, cont.

- Λ is the **attenuation matrix** or reliability matrix

$$\Lambda = \begin{pmatrix} \sigma_{zz} & \sigma_{zx} \\ \sigma_{xz} & \sigma_{xx} + \sigma_{uu} \end{pmatrix}^{-1} \begin{pmatrix} \sigma_{zz} & \sigma_{zx} \\ \sigma_{xz} & \sigma_{xx} \end{pmatrix}$$

- The terms σ_{zz}, etc. are now covariance matrices
- Biases can take many forms, including **reversal of signs**!
- **Global null test is OK**: Naive test of $H_0 : \beta_x = 0,$ and$\beta_z = 0$ is valid
- **No effect for any component of X is OK**: Naive test of $H_0 : \beta_x = 0$ is valid

What Naive Tests are Invalid?

- **Tests for Z**: surprisingly, tests about the variables Z measured **without error** are **not valid**
  - Exception is when X and Z are independent
- **When X is multivariate**: Naive tests here for components of X are **not valid**

Bias for Inferences About Error-Free Covariates

- **When X and Z are related**

- **Regression of X on Z**:

$$E(X \mid Z) = \Gamma_1 + \Gamma_z^{\mathsf{T}} Z$$

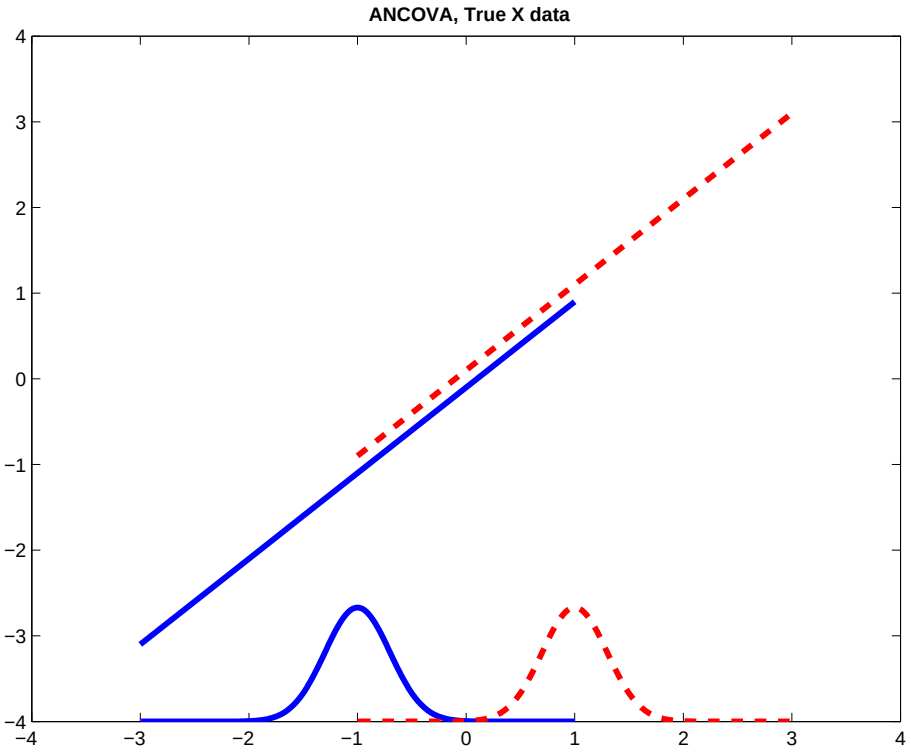
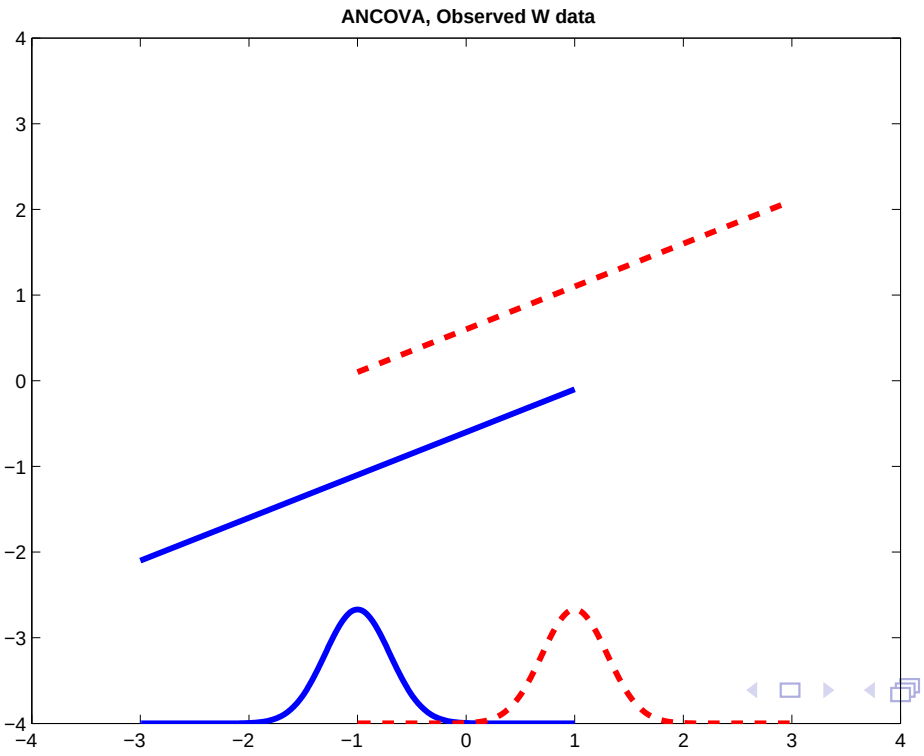
- **Effects of Error**: You do not estimate β_z, but instead you estimate

$$\beta_{z*} = \beta_z + (1 - \lambda_1)\beta_x \Gamma_z,$$

Analysis Of Covariance

- These results have implications for the **two group ANCOVA**.
- **True Predictor** X
- **Group Assignment** $Z =$ dummy indicator of group 1, say
- **Imbalance**: If X has a different mean for the two groups, then the estimated effect of Z is **biased**
- **Illustration**: The next slides illustrate that even when there is no Z effect in truth, the observed data may indicate, **falsely**, that there is an apparent effect

2-Group ANCOVA, True X Data. Note no effect.



2-Group ANCOVA, Observed W Data. Note apparent effect.